

Results in Engineering

Robustness-Integrated Evidence and Choice (RIEC)-L1: anauditable hybrid criterion for model selection in industrialcurve-fitting tasks (optical spectra, RTD, and thin-layer drying) --Manuscript Draft--

Manuscript Number:	RINENG-D-26-03644
Full Title:	Robustness-Integrated Evidence and Choice (RIEC)-L1: anauditable hybrid criterion for model selection in industrialcurve-fitting tasks (optical spectra, RTD, and thin-layer drying)
Short Title:	Many industrial and process-engineering tasks reduce to one-dimensional curve fitting and outof-condition prediction, yet candidate models span different families and there is no unified, auditable standard for deciding the necessary complexity—especially under deployment-like splits such as leave-one-condition-out. We define RIEC (Robustness-Integrated Evidence and Choice) and implement a minimal variant, RIEC-L1, for model selection over a finite library $\mathcal{M} = \{m_1, \dots, m_M\}$. RIEC-L1 combines a BIC-style structural penalty with task-aligned cross-validated evidence, using a baselinenormalized extrapolation improvement factor (XPE), to produce a single hybrid score $S_{RIEC-L1}$. We demonstrate the approach on three industrial case datasets: optical reflectance s
Article Type:	Research paper
Section/Category:	AI and ML Engineering
Keywords:	model selection cross-validation BIC reproducibility
Corresponding Author:	Jincheng Li University of Nottingham - China Campus Ningbo, Zhejiang CHINA
Corresponding Author Secondary Information:	
Corresponding Author's Institution:	University of Nottingham - China Campus
Corresponding Author's Secondary Institution:	
First Author:	Jincheng Li
First Author Secondary Information:	
Order of Authors:	Jincheng Li Yuchen Zhao Xifeng Li
Order of Authors Secondary Information:	
Abstract:	Many industrial and process-engineering tasks reduce to one-dimensional curve fitting and out of-condition prediction, yet candidate models span different families and there is no unified, auditable standard for deciding the necessary complexity—especially under deployment-like

	splits such as leave-one-condition-out. We define RIEC (Robustness-Integrated Evidence and Choice) and implement a minimal variant, RIEC-L1, for model selection over a finite library $\mathcal{M} = \{m_1, \dots, m_M\}$. RIEC-L1 combines a BIC-style structural penalty with task-aligned cross-validated evidence, using a baseline normalized extrapolation improvement factor (XPE), to produce a single hybrid score $\mathcal{S}_{\text{RIEC-L1}}(m)$. We demonstrate the approach on three industrial case datasets: optical reflectance spectra ($n = 29872$), residence time distribution modeling ($n = 10401$), and thin-layer drying kinetics ($n = 72$) under leave-one-condition-out extrapolation on a public drying dataset [4]. In the optics and RTD cases, BIC, CV, and RIEC agree on the selected models, indicating stable gains in high-sample or well-structured settings. In the drying case, CV favors a flexible 4-parameter model, whereas BIC/RIEC prefer a parsimonious 2-parameter family; Page and Weibull are nearly tied, so we report winners via a deterministic tie-break and provide the full evidence ledger. The method yields a traceable model-choice record (ranking ledger + holdout overlays) with out requiring specialist statistical expertise. Code and data (one-click reproduction package) are available at https://github.com/lijincheng494gmail/result-in-eg-reproduce, and Supplementary S0–S3 provide a reader guide, protocol, derivations, and domain linkage notes.
Additional Information:	
Question	Response
Results in Engineering is an open access journal which charges an Article Publishing Charge (APC) to cover the cost associated with the publication process. All articles published Open Access will be immediately and permanently free on ScienceDirect for users to read, download, and use in accordance with the author's selected Creative Commons user	Yes

license. As an Author, I acknowledge I need to pay the Article Publishing Charge if my manuscript is accepted for publication.	
To complete your submission you must select a statement which best reflects the availability of your research data/code. IMPORTANT: this statement will be published alongside your article. If you have selected "Other", the explanation text will be published verbatim in your article (online and in the PDF). (If you have not shared data/code and wish to do so, you can still return to Attach Files. Sharing or referencing research data and code helps other researchers to evaluate your findings, and increases trust in your article. Find a list of supported data repositories in Author Resources, including the free-to-use multidisciplinary open Mendeley Data Repository.)	Other (please explain: e.g. 'I have shared the link to my data/code at the Attach File step').
You have selected other. Please explain: as follow-up to "To complete your submission you must select a statement which best reflects the availability of your research data/code. IMPORTANT: this statement will be published alongside your article. If you have selected "Other", the explanation text will be published verbatim in your article (online and in the PDF). (If you have not shared data/code and wish to do so, you can still return to Attach Files. Sharing or referencing research data and code helps other researchers to evaluate your findings, and increases trust in your article. Find a list of supported data repositories in Author Resources, including the free-to-use multidisciplinary open Mendeley Data Repository.)"	Reproducibility package (code+data+scripts) is available at https://github.com/lijincheng494gmail/result-in-eg-reproduce (ZIP: riec_rieg_reproduce.zip). Drying dataset DOI: 10.17632/jdbkxgn2v5.1.
Free Preprint Service Do you want to share your research early as a preprint? Preprints allow for open access to and citations of your research prior to publication.	YES, I want to share my research early and openly as a preprint.

Results in Engineering offers a free service to post your paper on SSRN, an open access research repository, when your paper enters peer review. Once on SSRN, your paper will benefit from early registration with a DOI and early dissemination that facilitates collaboration and early citations. It will be available free to read regardless of the publication decision made by the journal. This will have no effect on the editorial process or outcome with the journal. Please consult the [SSRN Terms of Use](#) and [FAQs](#).

Cover Letter

Manuscript title: *RIEC-L1: an auditable hybrid criterion for industrial curve-model selection under hard extrapolation*

Corresponding author: Jincheng Li (University of Nottingham Ningbo China, UNNC)

Email: smyj12@nottingham.edu.cn

Date: February 24, 2026

Dear Editor,

We submit the enclosed manuscript proposing **RIEC-L1** (Robustness-Integrated Evidence and Choice, Layer 1), a minimal and *auditable* hybrid criterion that combines a BIC-style structural penalty with deployment-aligned out-of-fold evidence (group cross-validation summarized via XPE). The method is designed for industrial curve modeling workflows where engineers must select among a small library of physically meaningful candidate models and must produce a traceable model-choice record suitable for quality management and review.

Key contributions.

- A simple hybrid score C_λ that balances structural parsimony (BIC) and predictive robustness (CV/XPE), with a monotone weight schedule $\lambda(n_{\text{eff}})$.
- A deployment-aligned evaluation protocol (grouped CV by spectrum/run/condition) that targets hard extrapolation settings (e.g., leave-one-condition-out).
- Three industrial case studies (optical spectra, RTD, and thin-layer drying) demonstrating when BIC/CV agree and when the hybrid criterion provides a conservative compromise.
- An evidence-first reporting format: ranked ledgers and holdout overlays enabling non-specialists to audit model-choice decisions.

File list for submission (single-file uploads).

- **File 1:** Cover letter (this PDF).
- **File 2:** Manuscript.
- **File 3:** Figures Set (Fig. 1–7 + Section B evidence snapshot charts).
- **File 4:** Tables Set (Table 1 + per-case evidence tables).
- **File 5:** Minimal mathematical definitions (see Supplementary S2 for derivations).
- **File 6:** Reader Guide (also referred to as **Supplementary S0**).
- **File 7:** Reproducibility links and data/code availability statement.
- **Supplementary S1:** Protocol and reproducibility details (pipeline design; no code required for peer review).
- **Supplementary S2:** Formal definitions and derivations.
- **Supplementary S3:** Domain background and linkage notes.

Data and code availability. The drying dataset is public (DOI: 10.17632/jdbkxgn2v5.1). For peer review, we provide an auditable evidence snapshot (Files 3–4) and a standardized protocol description (Supplementary S1). Upon acceptance, a public, versioned repository release (code + data) will be made available under <https://github.com/lijincheng494gmail>.

Sincerely,

Jincheng Li

Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

- Hybrid score combines BIC with deployment-aligned CV evidence (XPE) for auditable selection.
- Demonstrated on three industrial case studies spanning large to small effective sample sizes.
- Under hard extrapolation (LOCO), RIEC prefers parsimonious models when pure CV favors complexity.
- Outputs a ranked evidence ledger and holdout overlays to support engineering deployment and review.

Robustness-Integrated Evidence and Choice (RIEC)-L1: an auditable hybrid criterion for model selection in industrial curve-fitting tasks (optical spectra, RTD, and thin-layer drying)

Jincheng Li^a, Yuchen Zhao^b and Xifeng Li^c

^aFaculty of Science and Technology, University of Nottingham Ningbo China (UNNC), 199 Taikang East Road, Yinzhou District, Ningbo, Zhejiang, 315100, China

^bDepartment of Chemistry, College of Chemistry and Chemical Engineering, Xiamen University, 422 Siming Road, Siming District, Xiamen, Fujian, 361005, China

^cIT Department, China National Cereals, Oils and Foodstuffs Corporation (COFCO), COFCO Fortune Plaza, No. 8 Chaoyangmen South Street, Chaoyang District, Beijing, 100020, China

ARTICLE INFO

Keywords:

model selection
cross-validation
BIC
industrial curve fitting
RTD
thin-layer drying
reproducibility

ABSTRACT

Many industrial and process-engineering tasks reduce to one-dimensional curve fitting and out-of-condition prediction, yet candidate models span different families and there is no unified, auditable standard for deciding the necessary complexity—especially under deployment-like splits such as leave-one-condition-out.

We define **RIEC (Robustness-Integrated Evidence and Choice)** and implement a minimal variant, **RIEC-L1**, for model selection over a finite library $M = (C, f)$. **RIEC-L1** combines a BIC-style structural penalty with task-aligned cross-validated evidence, using a baseline-normalized extrapolation improvement factor (XPE), to produce a single hybrid score C_λ .

We demonstrate the approach on three industrial case datasets: optical reflectance spectra ($n_{\text{eff}} = 29872$), residence time distribution modeling ($n_{\text{eff}} = 10401$), and thin-layer drying kinetics ($n_{\text{eff}} = 72$) under leave-one-condition-out extrapolation on a public drying dataset [4]. In the optics and RTD cases, BIC, CV, and RIEC agree on the selected models, indicating stable gains in high-sample or well-structured settings. In the drying case, CV favors a flexible 4-parameter model, whereas BIC/RIEC prefer a parsimonious 2-parameter family; Page and Weibull are nearly tied, so we report winners via a deterministic tie-break and provide the full evidence ledger.

The method yields a traceable model-choice record (ranking ledger + holdout overlays) without requiring specialist statistical expertise. Code and data (one-click reproduction package) are available at <https://github.com/lijincheng494gmail/result-in-eg-reproduce>, and Supplementary S0–S3 provide a reader guide, protocol, derivations, and domain linkage notes.

1. Introduction

Digital-transformation programs in large industrial groups often start from diagnosing and modeling critical processes or equipment, where field data frequently appear as one-dimensional curves over time or wavenumber (temperature traces, response curves, spectra, and so on). In such tasks, engineering teams must choose among multiple candidate models, including mechanistic descriptions, empirical curve families, and piecewise models. In practice, “more complex” often means more parameters, more degrees, or more segments, which can improve in-sample fit but may degrade extrapolation and interpretability. This paper focuses on an engineering question: *how to decide whether extra complexity is truly necessary, using evidence that is auditable and aligned with deployment.*

Why a hybrid criterion is needed. A major tension is that information criteria such as BIC are conservative in small samples, while cross-validation (CV) can favor flexible models when the evaluation split inadvertently leaks information or fails to reflect the true deployment setting. The relationship between information criteria and CV has been studied extensively: under regular conditions, leave-one-out CV is asymptotically equivalent to AIC [7], and in Bayesian settings information-criterion approximations such as WAIC track leave-one-out predictive evaluation [8, 9]. At the same time, CV can be inconsistent for model-dimension selection under common split regimes [10],

¹Corresponding author. Email: smyjl12@nottingham.edu.cn.

RIEC-L1: auditable hybrid criterion

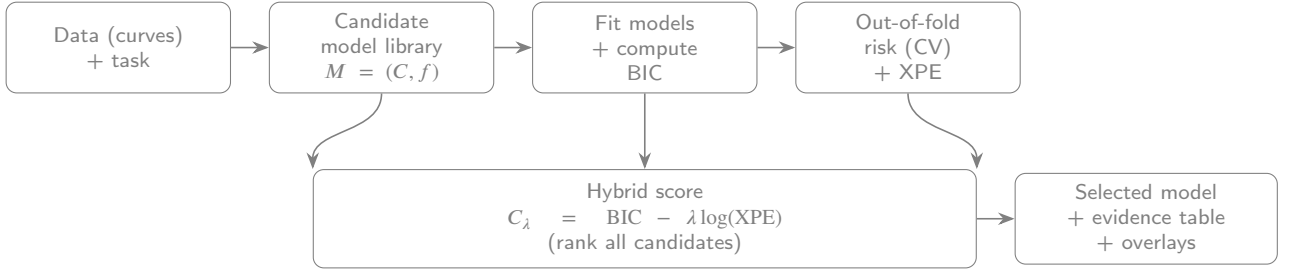


Figure 1: RIEC-L1 workflow. Candidate models are indexed as $M = (C, f)$ (structure C and complexity f). We combine a structural penalty (BIC) with out-of-fold evidence (CV and XPE) into a single auditable score C_λ and output a ranked ledger plus holdout overlays for inspection.

motivating careful, task-aligned grouping (by run, batch, or condition) rather than random shuffles. However, much of the existing work is oriented toward statistical theory or domain-specific settings, and it rarely targets industrial curve-fitting realities: *hard extrapolation* (e.g., leave-one-condition-out deployment), small physically interpretable candidate libraries (typically 5–10 models), and the need to produce an *auditable* decision record that non-specialists can review. We therefore implement **RIEC-L1**, a minimal layer of **RIEC** (Robustness-Integrated Evidence and Choice), which unifies structural penalties and out-of-fold evidence into one score and produces a single ranking table that can be audited by non-specialists.

Contributions and scope. This paper makes three practical contributions: (i) we define **RIEC** as a robustness-oriented evidence framework for engineering model choice and instantiate it as a minimal hybrid criterion (RIEC-L1) that integrates a BIC-style penalty with task-aligned out-of-fold evidence via XPE; (ii) we emphasize *auditable artifacts*—a ranked evidence ledger and holdout overlays—so that model-choice decisions can be reviewed and stored in industrial workflows; and (iii) we demonstrate the approach on three representative curve-fitting tasks (optics, RTD, drying) under deployment-like splits. The goal is not to propose new physical models, but to provide a lightweight, transparent *selection* layer that can sit on top of any candidate model library.

Definition (RIEC). **RIEC** (Robustness-Integrated Evidence and Choice) is an auditable evidence framework for engineering model choice. It integrates (i) *structural evidence* (complexity penalties such as BIC), (ii) *predictive evidence* from task-aligned out-of-fold evaluation (e.g. CV and XPE), and (iii) *decision artifacts* (ranking ledgers and holdout overlays) that can be inspected and stored. RIEC-L1 in this paper instantiates the framework as a single hybrid score C_λ over a finite library.

2. Methods: RIEC-L1 model selection with auditable evidence

We consider a finite candidate set $\mathcal{M} = \{M_1, \dots, M_{K_{\text{cand}}}\}$, where each model is indexed as $M = (C, f)$: C is a structural switch (worldview) and f is the complexity level within that structure. For each candidate, we fit parameters by maximum likelihood (or least squares under Gaussian noise) and obtain the maximized log-likelihood $\ell_n(\hat{\theta}_M; M)$ and parameter count k_M .

2.1. BIC-style structural penalty

We use the Bayesian information criterion (BIC) [1] with an effective sample size n_{eff} :

$$\text{BIC}(M) = -2\ell_n(\hat{\theta}_M; M) + k_M \log(n_{\text{eff}}). \quad (1)$$

Smaller values indicate a better trade-off between fit and complexity.

Effective sample size in curve tasks. In curve-fitting, point-wise residuals along a curve can be serially correlated, so the “sample size” entering classical BIC derivations (which assume independent observations) is ambiguous. RIEC-L1 is designed for engineering audibility, so we adopt a transparent proxy: we define n_{eff} as the number of scalar likelihood terms used in fitting (equivalently, the number of stacked (x, y) pairs after putting all curves into a long-form table under the chosen split). This definition is trivial to compute, requires no specialist time-series corrections,

and makes the strength of the $k_M \log(n_{\text{eff}})$ penalty explicit and auditable. For complete traceability, the three cases decompose as: Optics: 4 spectra $\times$ 7468 wavenumber samples = 29872; RTD: concatenating all runs yields 10401 ($t, E(t)$) samples; Drying: 6 (material, temperature) conditions $\times$ 12 time points = 72. This is a pragmatic balance between theoretical rigor and deployability: in strongly correlated settings one may substitute a block- or correlation-adjusted effective sample size, and the RIEC-L1 framework is unchanged.

2.2. Out-of-fold risk and improvement factor (XPE)

Let E_M^{CV} denote the K_{cv} -fold cross-validated risk [2, 3] under a task-aligned split (e.g., by run or by condition). Throughout, we use mean squared error (MSE) over held-out curve points as the risk: for each fold we compute the MSE on the held-out group (spectrum/run/condition) and then average across folds. No additional rescaling is applied beyond domain-standard normalizations already defined for each case (e.g., RTD kernel normalization and moisture-ratio curves).

We define an improvement factor relative to a baseline model M_{base} :

$$\text{XPE}(M) = \frac{E_{\text{base}}^{\text{CV}}}{E_M^{\text{CV}}}. \quad (2)$$

Values > 1 indicate better out-of-fold performance than the baseline.

Baseline choice. Unless stated otherwise, we take M_{base} to be the simplest candidate in the library (lowest parameter count within the case), which corresponds to a reasonable “engineering default”. When multiple candidates share the minimal parameter count, we choose a deterministic default for reproducibility (in the RTD case we use `gamma_2p` as the standard engineering RTD family). The chosen baseline is reported in the released ranking ledger (Supplementary S1). This makes XPE easy to interpret as the factor-of-improvement gained by adding complexity beyond a baseline model.

2.3. Hybrid criterion

RIEC-L1 uses a hybrid criterion

$$C_\lambda(M) = \text{BIC}(M) - \lambda(n_{\text{eff}}) \log(\text{XPE}(M)), \quad (3)$$

where $\lambda(n_{\text{eff}})$ decreases with sample size. In experiments we use

$$\lambda(n_{\text{eff}}) = \frac{1}{\log(n_{\text{eff}})}, \quad (4)$$

which corresponds to $c = 1$ in $\lambda = c / \log n_{\text{eff}}$ ($\log$ is the natural logarithm).

Rationale for $\lambda(n_{\text{eff}}) = c / \log n_{\text{eff}}$. The $1 / \log n_{\text{eff}}$ decay is a minimal monotone schedule that guarantees $\lambda \rightarrow 0$ as $n_{\text{eff}} \rightarrow \infty$, so C_λ asymptotically reduces to BIC and retains its large-sample behavior. Conversely, when n_{eff} is small, λ is larger and the out-of-fold term can meaningfully counterbalance BIC’s tendency to over-penalize complexity. We use $c = 1$ as a conservative default intended to work across typical industrial curve libraries; practitioners can tune c upward to prioritize predictive robustness or downward to prioritize structural parsimony. Across the three heterogeneous cases in Table 1, this default produces engineering-reasonable selections without case-specific tuning.

The selected model is $M^\star = \arg \min_{M \in \mathcal{M}} C_\lambda(M)$.

2.4. Standardized execution steps (auditable protocol)

For practical deployment, RIEC-L1 can be executed as the following compact, auditable procedure:

1. **Define the candidate library.** Specify the candidate set $\mathcal{M}$ as models $M = (C, f)$, and choose the simplest plausible model as a baseline M_{base} (for XPE normalization).
2. **Choose a deployment-aligned split.** Set the CV grouping rule to match the intended deployment boundary (e.g., leave-one-run-out, leave-one-batch-out, or leave-one-condition-out).
3. **Fit each candidate model.** For every $M \in \mathcal{M}$, fit parameters on the training data in each fold (maximum likelihood / least squares).

4. **Compute auditable evidence.** For each M , compute BIC, out-of-fold risk E_M^{CV} , and the improvement factor $\text{XPE}(M) = E_{\text{base}}^{\text{CV}} / E_M^{\text{CV}}$.
5. **Compute the hybrid score and rank.** Compute n_{eff} and $\lambda = c / \log n_{\text{eff}}$ (default $c = 1$), then score $C_\lambda(M)$ and rank candidates from low to high.
6. **Select and archive artifacts.** Choose M^* (lowest C_λ), and archive the full ranking ledger plus representative holdout overlays for inspection and quality/audit traceability.

3. Data, tasks, and evaluation protocol

This section specifies the data, tasks, and evaluation protocol for three cases, ensuring that out-of-fold evidence corresponds to realistic deployment settings. All experiments output (i) a candidate ranking table with BIC/CV/XPE/ C_λ and (ii) an overlay plot comparing holdout curves against fitted predictions. For peer review, the optics and RTD datasets used in this study are provided in the reproducibility materials (Supplementary S1); the drying dataset is public [4].

3.1. Case I: optical reflectance spectra

Data: four reflectance spectra (Si and SiC, each measured at two incidence angles), with wavenumber w (cm^{-1}) as input and reflectance $R(w)$ as output. **Task:** learn the spectral curve shape and extrapolate to unseen spectra (held-out material/angle condition). **Split:** leave-one-spectrum-out (each spectrum is a `condition_id`).

3.2. Case II: residence time distribution (RTD)

Data: multiple tracer-response runs sampled as outlet concentration $c(t)$. We form an empirical RTD kernel $E(t) = c(t) / \int c(t) dt$ [11, 12]. **Task:** select an interpretable RTD distribution family that extrapolates across runs. **Split:** leave-one-run-out (each run is a group; we hold out one full $E(t)$ curve).

3.3. Case III: thin-layer drying kinetics

Data: moisture-ratio curves $\text{MR}(t)$ for two rice types (Nadu and Samba) under three temperatures (45/55/65°C), extracted from a public spreadsheet [4]. **Task:** predict $\text{MR}(t)$ decay curves and extrapolate to unseen temperature–material conditions. **Split:** leave-one-condition-out (LOCO), where a condition is a (material, temperature) pair.

4. Case studies and model libraries

4.1. Case I: optical spectra (proxy family)

We use an auditable proxy family inspired by thin-film interference modeling [13, 14]:

$$R(w) = P_f(w) + A \cos(\omega w + \phi), \quad (5)$$

where P_f is an f -degree polynomial and (A, ω, ϕ) are sinusoidal parameters. Here C denotes the structure “polynomial + cosine” and $f \in \{0, 1, 2\}$ controls complexity via polynomial degree. We compare `cos_poly_deg0/1/2`.

4.2. Case II: RTD distribution families

We compare three classical RTD worldviews with matched parameter counts:

- **Gamma / tanks-in-series** (`gamma_2p`), a standard RTD family consistent with an equivalent CSTR-in-series interpretation [11, 12].
- **Lognormal** (`lognormal_2p`), representing multiplicative/long-tail effects.
- **Axial-dispersion proxy** (`axdisp_2p`), a two-parameter approximation motivated by longitudinal dispersion [15].

Because parameter counts are matched, this case emphasizes worldview comparison rather than parameter inflation.

4.3. Case III: thin-layer drying model families

We compare widely used thin-layer drying models [5, 6, 16]:

- **Lewis/Newton** (`lewis_1p`): $MR(t) = \exp(-kt)$.
- **Page** (`page_2p`): $MR(t) = \exp(-kt^n)$.
- **Weibull** (`weibull_2p`): $MR(t) = \exp(-(t/\alpha)^\beta)$.
- **Two-term exponential** (`two_term_3p`).
- **Midilli-Kucuk** (`midilli_4p`): $MR(t) = a \exp(-kt^n) + bt$.
- **Arrhenius-tied variants** (`arrhenius_lewis_2p`, `arrhenius_page_3p`): tie the rate constant across temperatures via an Arrhenius form; evaluated as an engineering extension (see Supplementary S1).

These Arrhenius-tied variants did not reduce LOCO out-of-condition risk in our dataset and are included for completeness in the evidence ledger (Supplementary S1).

Here C is the family choice and f is the complexity level as reflected by parameter count and structural flexibility.

5. Results

Table 1 summarizes effective sample size, the chosen weight λ , and the final selections under RIEC/BIC/CV for each case. Figures 2–7 provide holdout overlays and C_λ ranking evidence.

In the Optics (large n_{eff}) and RTD (moderate n_{eff}) cases, all three criteria agree, indicating that out-of-fold gains are stable and RIEC does not introduce extra selection volatility. In the Drying case, CV selects the more flexible `midilli_4p` model (4 parameters) for the smallest holdout error. BIC and RIEC both favor a parsimonious 2-parameter family under LOCO extrapolation; `page_2p` and `weibull_2p` yield nearly indistinguishable evidence (Supplementary S1), so we report `weibull_2p` as the BIC winner and `page_2p` as the RIEC representative under a deterministic tie-breaking rule.

Case I (Optics): stable selection under abundant samples. Figures 2–3 show that the selected `cos_poly_deg2` model matches the held-out spectrum well and dominates the hybrid score ranking. Because n_{eff} is large, $\lambda(n_{\text{eff}})$ is small and the RIEC-L1 score is close to BIC, yet the out-of-fold term still provides an interpretable sanity check ($\text{XPE} > 1$ indicates a measurable gain over the baseline). The narrow spread in C_λ across candidates reflects selection stability: small modeling changes do not flip the ranking.

Case II (RTD): worldview comparison with matched complexity. For RTD, candidates have matched parameter counts, so ranking mainly reflects which *distribution family* best generalizes across runs. Figures 4–5 show that `gamma_2p` provides the best cross-run extrapolation and the lowest C_λ . In this setting, RIEC-L1 preserves the standard engineering conclusion (a tanks-in-series/gamma RTD) while still producing a reusable evidence record. The fitted gamma parameters map directly to mean residence time and dispersion (or an equivalent number of tanks), supporting interpretability for process engineers.

Case III (Drying, LOCO): conservative choice under hard extrapolation. Drying under LOCO split is intentionally challenging: each held-out fold removes an entire (material, temperature) condition, so flexible models can overfit small idiosyncrasies. Figures 6–7 highlight the key disagreement: CV prefers `midilli_4p`, while BIC prefers a simpler 2-parameter family. RIEC-L1 selects `page_2p` because its out-of-fold gain is substantial (high XPE) yet does not require the full flexibility of a 4-parameter form. In this case, `page_2p` and `weibull_2p` are essentially tied in evidence (Supplementary S1 ledger), so the specific 2-parameter representative is chosen by a deterministic tie-break; the substantive distinction is the conservative preference for 2 parameters over 4 under hard extrapolation. Quantitatively, `midilli_4p` lowers the CV risk from 4.15×10^{-3} (`page_2p`) to 4.00×10^{-3} (about a 3.5% improvement), but it doubles the parameter count (4 vs 2) and is penalized by the structural term (BIC increases by $\Delta \text{BIC} \approx 4.6$ relative to `page_2p`). RIEC-L1 therefore accepts a modest loss in CV fit to halve model complexity and improve robustness and interpretability under hard out-of-condition deployment. This illustrates the intended policy of RIEC: *do not add complexity unless the out-of-fold evidence is clearly worth it.*

Table 1

Three-case summary. Lower is better for BIC and C_λ . XPE > 1 indicates out-of-fold improvement over baseline. In the drying case, page_2p and weibull_2p are nearly tied at the reported precision; the reported winners follow a deterministic tie-break (see the full ledger in Supplementary S1).

Case	RIEC best	BIC best	CV best	n_{eff}	λ	XPE (RIEC)
Optics	cos_poly_deg2	cos_poly_deg2	cos_poly_deg2	29872	0.097	1.035
RTD	gamma_2p	gamma_2p	gamma_2p	10401	0.108	1.000
Drying	page_2p	weibull_2p	midilli_4p	72	0.234	1.569

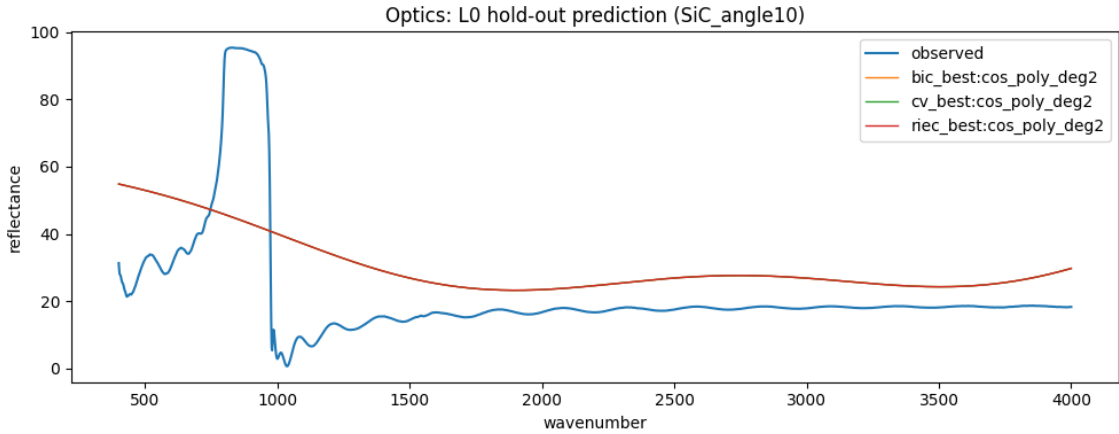


Figure 2: Case I (Optics): holdout overlay. Leave-one-spectrum-out holdout overlay for an unseen spectrum (SiC at 10° incidence; representative condition). The blue curve is the observed reflectance $R(w)$ on the held-out spectrum; the colored curves are predictions from the models chosen by BIC, task-aligned CV, and RIEC-L1. In this large- n_{eff} case the three selectors coincide on cos_poly_deg2, hence the predicted curves overlap. The overlay is an auditable check that the selected complexity captures the main spectral trend without introducing visually obvious overfitting artifacts.

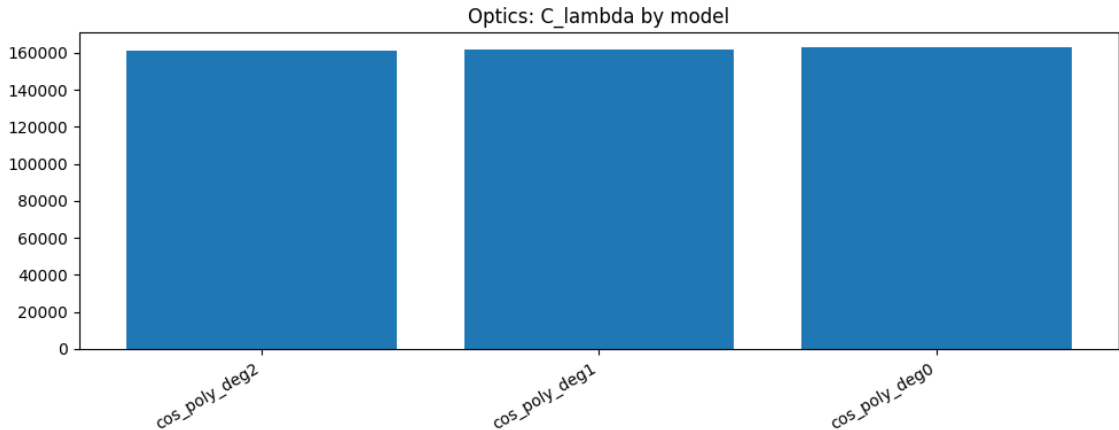


Figure 3: Case I (Optics): C_λ ranking evidence. Candidate ranking by the hybrid score C_λ across the proxy library $\{\text{cos_poly_deg0/1/2}\}$ (lower is better). The minimum at cos_poly_deg2 matches the selection in Table 1 and is supported by a small but consistent out-of-fold improvement (XPE > 1) while the BIC penalty remains modest. The tight spread of scores indicates selection stability in the large- n_{eff} regime.

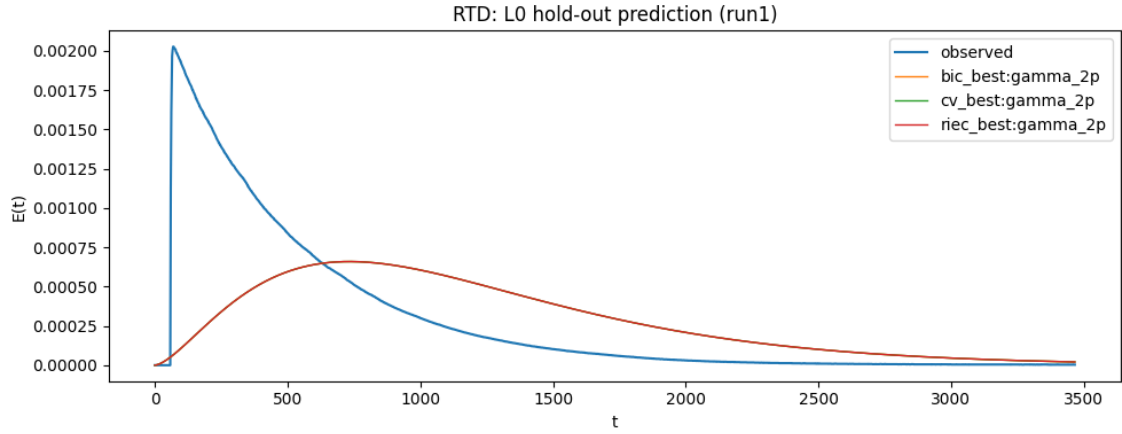


Figure 4: Case II (RTD): holdout overlay. Leave-one-run-out holdout overlay for an unseen tracer-response run (representative example). We plot the empirical RTD kernel $E(t)$ (blue) together with predictions from the models selected by BIC, CV, and RIEC-L1. In this moderate- n_{eff} case all three selectors choose gamma_2p , hence the predicted curves coincide. This demonstrates that the hybrid criterion preserves the engineering conclusion when evidence is strong and the candidate worldviews are well separated.

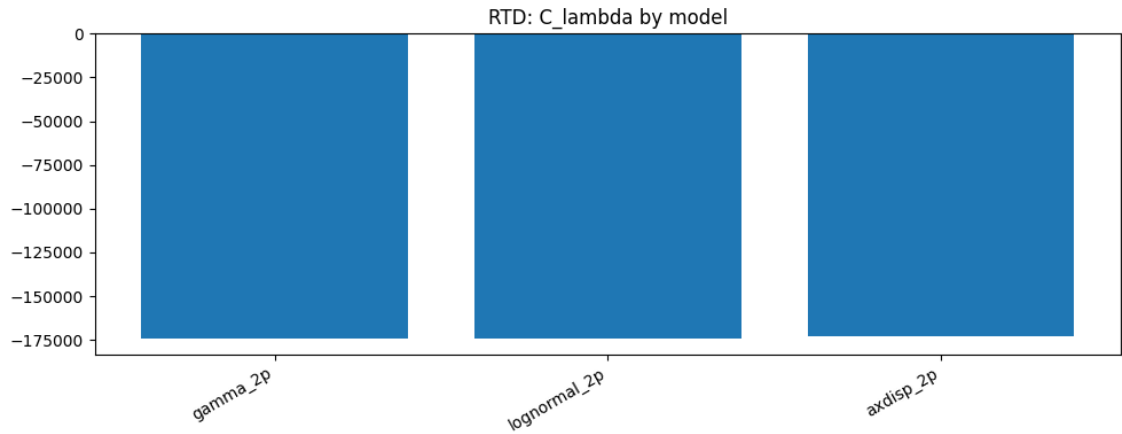


Figure 5: Case II (RTD): C_λ ranking evidence. Candidate ranking by C_λ over three competing RTD families (gamma_2p , lognormal_2p , axdisp_2p). Because parameter counts are matched, this ranking primarily reflects worldview adequacy under out-of-fold evaluation. gamma_2p achieves the lowest score, consistent with its interpretable tanks-in-series form and stable generalization across runs.

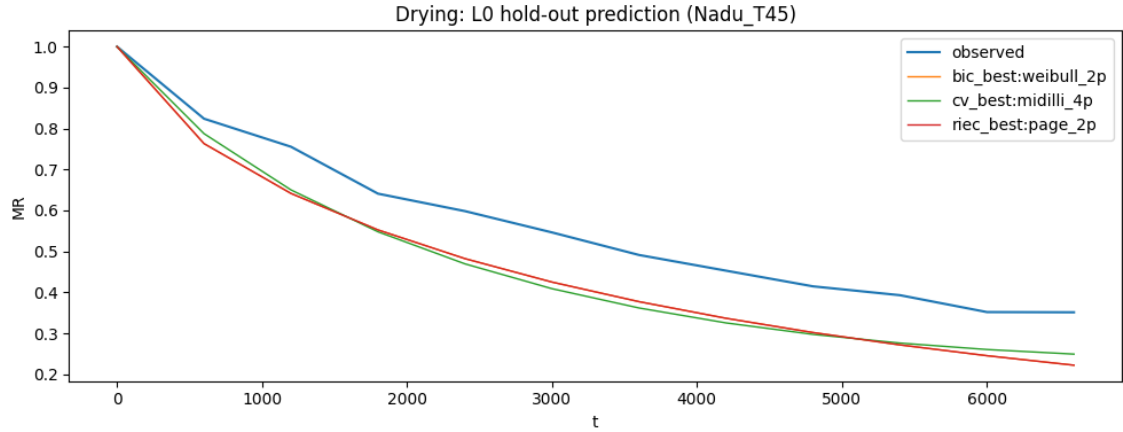


Figure 6: Case III (Drying): holdout overlay under LOCO extrapolation. Leave-one-condition-out (material, temperature) holdout overlay for an unseen condition (Nadu at 45°C; representative). The flexible Midilli model achieves the lowest CV error (midilli_4p, green), whereas BIC favors a different 2-parameter family (weibull_2p, orange); RIEC-L1 selects the 2-parameter Page model (page_2p, red) as a robustness-oriented compromise. Visually, the RIEC choice tracks the holdout decay curve well while avoiding the extra degrees of freedom that may overfit in small-sample extrapolation.

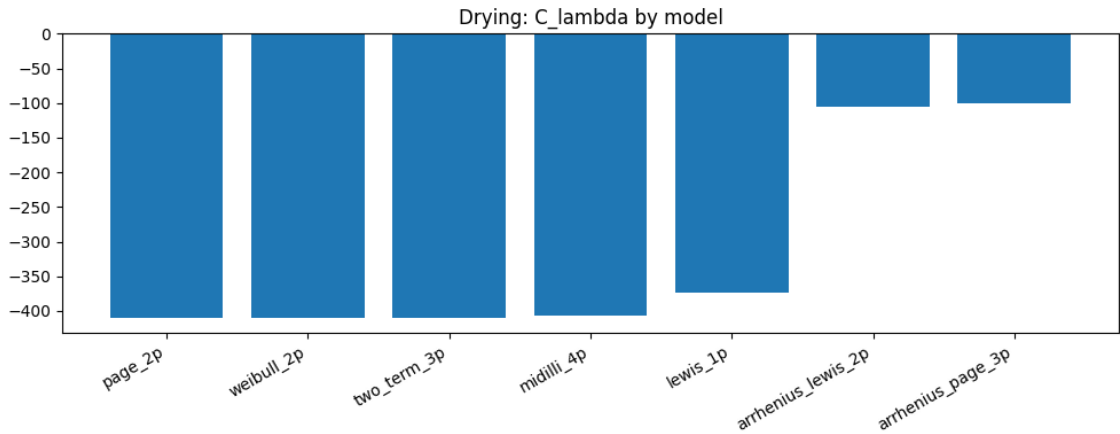


Figure 7: Case III (Drying): C_λ ranking evidence. Candidate ranking by C_λ across drying model families (lower is better). The hybrid score penalizes additional flexibility (e.g., midilli_4p) unless its out-of-fold gains are large enough to outweigh complexity, leading to the selection of page_2p. Together with Fig. 6, this ranking illustrates how RIEC-L1 operationalizes a conservative “do not add complexity unless necessary” policy under LOCO extrapolation.

6. Discussion

Across three cases, we address a practical question: in industrial curve tasks, is “more complex” truly necessary, and how can we decide using auditable evidence? When sample sizes are large or structural worldviews are clearly separated (Optics, RTD), RIEC agrees with BIC/CV and does not change the engineering conclusion. When sample sizes are small and extrapolation is harder (Drying, LOCO), CV can reward flexible models for marginal holdout gains; RIEC provides a conservative compromise that prioritizes parsimony when improvements are modest. In deployment, this reduces the risk of parameter inflation that complicates interpretation, tuning, and transfer to new conditions.

Practical usage in engineering workflows. RIEC is designed to be used as a lightweight “evidence gate” before deployment. In practice, we recommend (i) choosing a split that mirrors deployment (by run, by condition, or by batch), (ii) reporting the full ranking ledger (not only the winner), and (iii) inspecting at least one holdout overlay per group as a human-auditable check. When BIC and CV agree, the decision is typically stable; when they disagree, the ranking table provides a concrete starting point for engineering discussion (e.g., whether interpretability, stability, or accuracy is the dominant requirement). Beyond the three case studies here, the same protocol applies to many curve-like industrial tasks, including chemical process temperature–pressure trajectories, battery charge/discharge curves, motor vibration envelopes, and pharmaceutical dissolution profiles, where deployment often involves extrapolation across batches or operating conditions.

Limitations and future work. This paper intentionally uses a minimal Layer-1 instantiation. First, n_{eff} is treated as a transparent scalar-sample count to balance engineering usability and theoretical rigor; in curve-fitting the residuals can be serially correlated, so more refined effective-sample corrections (e.g., block-based or correlation-adjusted n_{eff}) may be beneficial in strongly dependent settings. Importantly, RIEC-L1 is compatible with any alternative n_{eff} definition because it only enters through $\log(n_{\text{eff}})$. Second, the current $\lambda(n_{\text{eff}}) = 1/\log n_{\text{eff}}$ is a simple monotone schedule; future layers could learn or calibrate λ for specific application families, or incorporate domain priors (e.g., physical constraints) as additional evidence channels. Finally, we focus on small model libraries; extending the approach to large hypothesis spaces (e.g., automated basis construction) is a natural direction.

7. Conclusion

We implement RIEC-L1 for industrial curve-fitting tasks: over a finite model library $M = (C, f)$, the hybrid score C_λ unifies BIC-style penalties with out-of-fold evidence (CV risk and XPE) into one auditable ranking table. For optical spectra and RTD, RIEC agrees with BIC/CV. For thin-layer drying, RIEC moderates the tendency of CV to favor the most flexible model in small-sample extrapolation. In industrial deployment, this can reduce overfitting-driven extrapolation failures, lower model maintenance cost, and provide a traceable model-selection record that supports engineering review and quality management. Supplementary materials provide a reader-oriented guide (S0), a one-click reproducibility protocol plus materials and an output snapshot for quick inspection (S1), formal definitions and derivations (S2), and domain background with engineering linkage notes (S3).

Declarations

CRedit authorship contribution statement

Jincheng Li: Conceptualization; Methodology; Software; Formal analysis; Visualization; Writing – original draft. Yuchen Zhao: Writing – review & editing (domain-specific mathematical modeling checks). Xifeng Li: Resources; Writing – review & editing (domain-specific mathematical modeling checks).

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare no competing interests.

Data and code availability

Code, data, and a cleaned one-click reproducibility package are publicly available at <https://github.com/lijincheng494gmail/result-in-eg-reproduce> (see also the reviewer-friendly archive `riec_rieg_reproduce.zip` referenced in File 7). For convenience, this submission also provides reviewer-oriented evidence snapshots (Figures: File 3; Tables: File 4) so the selection can be audited without running code. Supplementary S0–S3 provide a reader guide (S0), protocol description (S1), formal definitions/derivations (S2), and domain background/linkage notes (S3).

Declaration of generative AI and AI-assisted technologies

Generative AI tools were used for English-language proofreading and to assist in preparing the final submission files (e.g., formatting and consistency checks). All scientific content, analyses, and conclusions were produced and verified by the authors.

References

- [1] G. E. Schwarz, Estimating the dimension of a model, *Annals of Statistics* 6 (2) (1978) 461–464. doi:10.1214/aos/1176344136.
- [2] M. Stone, Cross-validatory choice and assessment of statistical predictions, *Journal of the Royal Statistical Society: Series B (Methodological)* 36 (2) (1974) 111–147.
- [3] S. Arlot, A. Celisse, A survey of cross-validation procedures for model selection, *Statistics Surveys* 4 (2010) 40–79. doi:10.1214/09-SS054.
- [4] C. Maduwanthi, Chathurika maduwanthi (dataset), *Mendeley Data*, V1 (2021). doi:10.17632/jdbkxgn2v5.1.
- [5] A. Midilli, H. Kucuk, Z. Yapar, A new model for single-layer drying, *Drying Technology* 20 (7) (2002) 1503–1513. doi:10.1081/DRT-120005864.
- [6] G. E. Page, Factors influencing the maximum rates of air drying shelled corn in thin layers, M.s. thesis, Purdue University (1949).
- [7] M. Stone, An asymptotic equivalence of choice of model by cross-validation and Akaike’s criterion, *Journal of the Royal Statistical Society: Series B (Methodological)* 39 (1) (1977) 44–47.
- [8] S. Watanabe, Asymptotic equivalence of Bayes cross validation and widely applicable information criterion in singular learning theory, *Journal of Machine Learning Research* 11 (2010) 3571–3594. URL <https://www.jmlr.org/papers/v11/watanabe10a.html>.
- [9] A. Vehtari, A. Gelman, J. Gabry, Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC, *Statistics and Computing* 27 (5) (2017) 1413–1432. doi:10.1007/s11222-016-9696-4.
- [10] J. Shao, Linear model selection by cross-validation, *Journal of the American Statistical Association* 88 (422) (1993) 486–494. doi:10.1080/01621459.1993.10476299.
- [11] P. V. Danckwerts, Continuous flow systems: Distribution of residence times, *Chemical Engineering Science* 2 (1) (1953) 1–13. doi:10.1016/0009-2509(53)80001-1.
- [12] O. Levenspiel, *Chemical Reaction Engineering*, 3rd Edition, Wiley, New York, 1999.
- [13] M. C. Tropicovsky, A. S. Sabau, A. R. Lupini, Z. Zhang, Transfer-matrix formalism for the calculation of optical response in multilayer systems: from coherent to incoherent interference, *Optics Express* 18 (24) (2010) 24715–24721. doi:10.1364/OE.18.024715.
- [14] O. S. Heavens, *Optical Properties of Thin Solid Films*, Dover Publications, New York, 1991.
- [15] O. Levenspiel, W. K. Smith, Notes on the diffusion-type model for the longitudinal mixing of fluids in flow, *Chemical Engineering Science* 6 (4–5) (1957) 227–233. doi:10.1016/0009-2509(57)85021-0.
- [16] S. M. Henderson, S. Pabis, Grain drying theory i: Temperature effect on drying coefficient, *Journal of Agricultural Engineering Research* 6 (1961) 169–174.

Robustness-Integrated Evidence and Choice (RIEC)-L1: an auditable hybrid criterion for model selection in industrial curve-fitting tasks (optical spectra, RTD, and thin-layer drying)

Jincheng Li^a, Yuchen Zhao^b and Xifeng Li^c

^aFaculty of Science and Technology, University of Nottingham Ningbo China (UNNC), 199 Taikang East Road, Yinzhou District, Ningbo, Zhejiang, 315100, China

^bDepartment of Chemistry, College of Chemistry and Chemical Engineering, Xiamen University, 422 Siming Road, Siming District, Xiamen, Fujian, 361005, China

^cIT Department, China National Cereals, Oils and Foodstuffs Corporation (COFCO), COFCO Fortune Plaza, No. 8 Chaoyangmen South Street, Chaoyang District, Beijing, 100020, China

ARTICLE INFO

Keywords:

model selection
cross-validation
BIC
industrial curve fitting
RTD
thin-layer drying
reproducibility

ABSTRACT

Many industrial and process-engineering tasks reduce to one-dimensional curve fitting and out-of-condition prediction, yet candidate models often come from different worldviews (i.e., model families / structural assumptions; mechanistic, empirical, or piecewise) and lack a unified, auditable evidence standard for complexity selection. We propose **RIEC (Robustness-Integrated Evidence and Choice)** and implement a minimal Layer-1 variant, **RIEC-L1**, for practical model selection over a finite model library $M = (C, f)$. RIEC-L1 combines a BIC-style structural penalty [1] with out-of-fold evidence from task-aligned cross-validation (CV) [2, 3], including a baseline-normalized improvement factor (XPE), to decide the *necessary* model complexity. We evaluate the approach on three industrial case datasets: optical reflectance spectra fitting ($n_{\text{eff}} = 29872$), residence time distribution (RTD) modeling ($n_{\text{eff}} = 10401$), and thin-layer drying kinetics ($n_{\text{eff}} = 72$, leave-one-condition-out extrapolation) based on a public drying dataset [4]. For the optics and RTD cases, BIC, CV, and RIEC agree on the selected models (`cos_poly_deg2` and `gamma_2p`), indicating stable gains in high-sample or well-structured settings. For the drying case, CV favors the more flexible `midilli_4p` model [5]; among the 2-parameter candidates, `page_2p` [6] and `weibull_2p` are nearly tied, so we report `weibull_2p` as the BIC winner and `page_2p` as the RIEC representative under a deterministic tie-breaking rule (full ledger in Supplementary S1). The criterion can be embedded as a standardized gate in digital twins, process modeling, and equipment diagnostics: it produces a traceable model-choice record without requiring specialist statistical expertise, reducing deployment and compliance friction. We release a reader-oriented guide (Supplementary S0), a one-click reproduction protocol plus materials and output snapshot (Supplementary S1), formal definitions and derivations (Supplementary S2), and domain background notes with engineering linkage (Supplementary S3) to support transparent model selection.

1. Introduction

Digital-transformation programs in large industrial groups often start from diagnosing and modeling critical processes or equipment, where field data frequently appear as one-dimensional curves over time or wavenumber (temperature traces, response curves, spectra, and so on). In such tasks, engineering teams must choose among multiple candidate models, including mechanistic descriptions, empirical curve families, and piecewise models. In practice, “more complex” often means more parameters, more degrees, or more segments, which can improve in-sample fit but may degrade extrapolation and interpretability. This paper focuses on an engineering question: *how to decide whether extra complexity is truly necessary, using evidence that is auditable and aligned with deployment.*

Why a hybrid criterion is needed. A major tension is that information criteria such as BIC are conservative in small samples, while cross-validation (CV) can favor flexible models when the evaluation split inadvertently leaks information or fails to reflect the true deployment setting. The relationship between information criteria and CV has been studied extensively: under regular conditions, leave-one-out CV is asymptotically equivalent to AIC [7], and in Bayesian settings information-criterion approximations such as WAIC track leave-one-out predictive evaluation

¹Corresponding author. Email: smyj112@nottingham.edu.cn.

RIEC-L1: auditable hybrid criterion

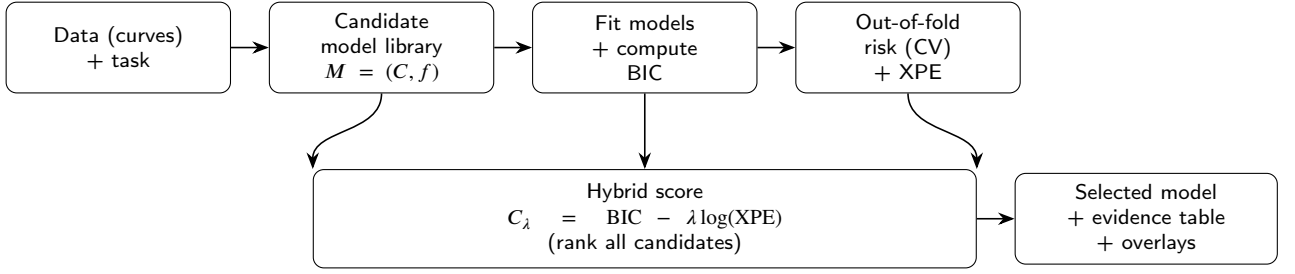


Figure 1: RIEC-L1 workflow. Candidate models are indexed as $M = (C, f)$ (structure C and complexity f). We combine a structural penalty (BIC) with out-of-fold evidence (CV and XPE) into a single auditable score C_λ and output a ranked ledger plus holdout overlays for inspection.

[8, 9]. At the same time, CV can be inconsistent for model-dimension selection under common split regimes [10], motivating careful, task-aligned grouping (by run, batch, or condition) rather than random shuffles. However, much of the existing work is oriented toward statistical theory or domain-specific settings, and it rarely targets industrial curve-fitting realities: *hard extrapolation* (e.g., leave-one-condition-out deployment), small physically interpretable candidate libraries (typically 5–10 models), and the need to produce an *auditable* decision record that non-specialists can review. We therefore implement **RIEC-L1**, a minimal layer of **RIEC** (Robustness-Integrated Evidence and Choice), which unifies structural penalties and out-of-fold evidence into one score and produces a single ranking table that can be audited by non-specialists.

Contributions and scope. This paper makes three practical contributions: (i) we define **RIEC** as a robustness-oriented evidence framework for engineering model choice and instantiate it as a minimal hybrid criterion (RIEC-L1) that integrates a BIC-style penalty with task-aligned out-of-fold evidence via XPE; (ii) we emphasize *auditable artifacts*—a ranked evidence ledger and holdout overlays—so that model-choice decisions can be reviewed and stored in industrial workflows; and (iii) we demonstrate the approach on three representative curve-fitting tasks (optics, RTD, drying) under deployment-like splits. The goal is not to propose new physical models, but to provide a lightweight, transparent *selection* layer that can sit on top of any candidate model library.

Definition (RIEC). **RIEC** (Robustness-Integrated Evidence and Choice) is an auditable evidence framework for engineering model choice. It integrates (i) *structural evidence* (complexity penalties such as BIC), (ii) *predictive evidence* from task-aligned out-of-fold evaluation (e.g. CV and XPE), and (iii) *decision artifacts* (ranking ledgers and holdout overlays) that can be inspected and stored. RIEC-L1 in this paper instantiates the framework as a single hybrid score C_λ over a finite library.

2. Methods: RIEC-L1 model selection with auditable evidence

We consider a finite candidate set $\mathcal{M} = \{M_1, \dots, M_{K_{\text{cand}}}\}$, where each model is indexed as $M = (C, f)$: C is a structural switch (worldview) and f is the complexity level within that structure. For each candidate, we fit parameters by maximum likelihood (or least squares under Gaussian noise) and obtain the maximized log-likelihood $\ell_n(\hat{\theta}_M; M)$ and parameter count k_M .

2.1. BIC-style structural penalty

We use the Bayesian information criterion (BIC) [1] with an effective sample size n_{eff} :

$$\text{BIC}(M) = -2\ell_n(\hat{\theta}_M; M) + k_M \log(n_{\text{eff}}). \quad (1)$$

Smaller values indicate a better trade-off between fit and complexity.

Effective sample size in curve tasks. In curve-fitting, point-wise residuals along a curve can be serially correlated, so the “sample size” entering classical BIC derivations (which assume independent observations) is ambiguous. RIEC-L1 is designed for engineering audibility, so we adopt a transparent proxy: we define n_{eff} as the number of scalar

likelihood terms used in fitting (equivalently, the number of stacked (x, y) pairs after putting all curves into a long-form table under the chosen split). This definition is trivial to compute, requires no specialist time-series corrections, and makes the strength of the $k_M \log(n_{\text{eff}})$ penalty explicit and auditable. For complete traceability, the three cases decompose as: Optics: 4 spectra $\times$ 7468 wavenumber samples = 29872; RTD: concatenating all runs yields 10401 $(t, E(t))$ samples; Drying: 6 (material, temperature) conditions $\times$ 12 time points = 72. This is a pragmatic balance between theoretical rigor and deployability: in strongly correlated settings one may substitute a block- or correlation-adjusted effective sample size, and the RIEC-L1 framework is unchanged.

2.2. Out-of-fold risk and improvement factor (XPE)

Let E_M^{CV} denote the K_{cv} -fold cross-validated risk [2, 3] under a task-aligned split (e.g., by run or by condition). Throughout, we use mean squared error (MSE) over held-out curve points as the risk: for each fold we compute the MSE on the held-out group (spectrum/run/condition) and then average across folds. No additional rescaling is applied beyond domain-standard normalizations already defined for each case (e.g., RTD kernel normalization and moisture-ratio curves).

We define an improvement factor relative to a baseline model M_{base} :

$$\text{XPE}(M) = \frac{E_{\text{base}}^{\text{CV}}}{E_M^{\text{CV}}}. \quad (2)$$

Values > 1 indicate better out-of-fold performance than the baseline.

Baseline choice. Unless stated otherwise, we take M_{base} to be the simplest candidate in the library (lowest parameter count within the case), which corresponds to a reasonable “engineering default”. When multiple candidates share the minimal parameter count, we choose a deterministic default for reproducibility (in the RTD case we use `gamma_2p` as the standard engineering RTD family). The chosen baseline is reported in the released ranking ledger (Supplementary S1). This makes XPE easy to interpret as the factor-of-improvement gained by adding complexity beyond a baseline model.

2.3. Hybrid criterion

RIEC-L1 uses a hybrid criterion

$$C_\lambda(M) = \text{BIC}(M) - \lambda(n_{\text{eff}}) \log(\text{XPE}(M)), \quad (3)$$

where $\lambda(n_{\text{eff}})$ decreases with sample size. In experiments we use

$$\lambda(n_{\text{eff}}) = \frac{1}{\log(n_{\text{eff}})}, \quad (4)$$

which corresponds to $c = 1$ in $\lambda = c / \log n_{\text{eff}}$ ($\log$ is the natural logarithm).

Rationale for $\lambda(n_{\text{eff}}) = c / \log n_{\text{eff}}$. The $1 / \log n_{\text{eff}}$ decay is a minimal monotone schedule that guarantees $\lambda \rightarrow 0$ as $n_{\text{eff}} \rightarrow \infty$, so C_λ asymptotically reduces to BIC and retains its large-sample behavior. Conversely, when n_{eff} is small, λ is larger and the out-of-fold term can meaningfully counterbalance BIC’s tendency to over-penalize complexity. We use $c = 1$ as a conservative default intended to work across typical industrial curve libraries; practitioners can tune c upward to prioritize predictive robustness or downward to prioritize structural parsimony. Across the three heterogeneous cases in Table 1, this default produces engineering-reasonable selections without case-specific tuning.

The selected model is $M^\star = \arg \min_{M \in \mathcal{M}} C_\lambda(M)$.

2.4. Standardized execution steps (auditable protocol)

For practical deployment, RIEC-L1 can be executed as the following compact, auditable procedure:

1. **Define the candidate library.** Specify the candidate set $\mathcal{M}$ as models $M = (C, f)$, and choose the simplest plausible model as a baseline M_{base} (for XPE normalization).
2. **Choose a deployment-aligned split.** Set the CV grouping rule to match the intended deployment boundary (e.g., leave-one-run-out, leave-one-batch-out, or leave-one-condition-out).

3. **Fit each candidate model.** For every $M \in \mathcal{M}$, fit parameters on the training data in each fold (maximum likelihood / least squares).
4. **Compute auditable evidence.** For each M , compute BIC, out-of-fold risk E_M^{CV} , and the improvement factor $\text{XPE}(M) = E_{\text{base}}^{\text{CV}} / E_M^{\text{CV}}$.
5. **Compute the hybrid score and rank.** Compute n_{eff} and $\lambda = c / \log n_{\text{eff}}$ (default $c = 1$), then score $C_\lambda(M)$ and rank candidates from low to high.
6. **Select and archive artifacts.** Choose M^* (lowest C_λ), and archive the full ranking ledger plus representative holdout overlays for inspection and quality/audit traceability.

3. Data, tasks, and evaluation protocol

This section specifies the data, tasks, and evaluation protocol for three cases, ensuring that out-of-fold evidence corresponds to realistic deployment settings. All experiments output (i) a candidate ranking table with BIC/CV/XPE/ C_λ and (ii) an overlay plot comparing holdout curves against fitted predictions. For peer review, the optics and RTD datasets used in this study are provided in the reproducibility materials (Supplementary S1); the drying dataset is public [4].

3.1. Case I: optical reflectance spectra

Data: four reflectance spectra (Si and SiC, each measured at two incidence angles), with wavenumber w (cm^{-1}) as input and reflectance $R(w)$ as output. **Task:** learn the spectral curve shape and extrapolate to unseen spectra (held-out material/angle condition). **Split:** leave-one-spectrum-out (each spectrum is a `condition_id`).

3.2. Case II: residence time distribution (RTD)

Data: multiple tracer-response runs sampled as outlet concentration $c(t)$. We form an empirical RTD kernel $E(t) = c(t) / \int c(t) dt$ [11, 12]. **Task:** select an interpretable RTD distribution family that extrapolates across runs. **Split:** leave-one-run-out (each run is a group; we hold out one full $E(t)$ curve).

3.3. Case III: thin-layer drying kinetics

Data: moisture-ratio curves $\text{MR}(t)$ for two rice types (Nadu and Samba) under three temperatures (45/55/65°C), extracted from a public spreadsheet [4]. **Task:** predict $\text{MR}(t)$ decay curves and extrapolate to unseen temperature–material conditions. **Split:** leave-one-condition-out (LOCO), where a condition is a (material, temperature) pair.

4. Case studies and model libraries

4.1. Case I: optical spectra (proxy family)

We use an auditable proxy family inspired by thin-film interference modeling [13, 14]:

$$R(w) = P_f(w) + A \cos(\omega w + \phi), \quad (5)$$

where P_f is an f -degree polynomial and (A, ω, ϕ) are sinusoidal parameters. Here C denotes the structure “polynomial + cosine” and $f \in \{0, 1, 2\}$ controls complexity via polynomial degree. We compare `cos_poly_deg0/1/2`.

4.2. Case II: RTD distribution families

We compare three classical RTD worldviews with matched parameter counts:

- **Gamma / tanks-in-series** (`gamma_2p`), a standard RTD family consistent with an equivalent CSTR-in-series interpretation [11, 12].
- **Lognormal** (`lognormal_2p`), representing multiplicative/long-tail effects.
- **Axial-dispersion proxy** (`axdisp_2p`), a two-parameter approximation motivated by longitudinal dispersion [15].

Because parameter counts are matched, this case emphasizes worldview comparison rather than parameter inflation.

4.3. Case III: thin-layer drying model families

We compare widely used thin-layer drying models [5, 6, 16]:

- **Lewis/Newton** (`lewis_1p`): $MR(t) = \exp(-kt)$.
- **Page** (`page_2p`): $MR(t) = \exp(-kt^n)$.
- **Weibull** (`weibull_2p`): $MR(t) = \exp(-(t/\alpha)^\beta)$.
- **Two-term exponential** (`two_term_3p`).
- **Midilli-Kucuk** (`midilli_4p`): $MR(t) = a \exp(-kt^n) + bt$.

Here C is the family choice and f is the complexity level as reflected by parameter count and structural flexibility.

5. Results

Table 1 summarizes effective sample size, the chosen weight λ , and the final selections under RIEC/BIC/CV for each case. Figures 2–7 provide holdout overlays and C_λ ranking evidence.

In the Optics (large n_{eff}) and RTD (moderate n_{eff}) cases, all three criteria agree, indicating that out-of-fold gains are stable and RIEC does not introduce extra selection volatility. In the Drying case, CV selects the more flexible `midilli_4p` model (4 parameters) for the smallest holdout error. BIC and RIEC both favor a parsimonious 2-parameter family under LOCO extrapolation; `page_2p` and `weibull_2p` yield nearly indistinguishable evidence (Supplementary S1), so we report `weibull_2p` as the BIC winner and `page_2p` as the RIEC representative under a deterministic tie-breaking rule.

Case I (Optics): stable selection under abundant samples. Figures 2–3 show that the selected `cos_poly_deg2` model matches the held-out spectrum well and dominates the hybrid score ranking. Because n_{eff} is large, $\lambda(n_{\text{eff}})$ is small and the RIEC-L1 score is close to BIC, yet the out-of-fold term still provides an interpretable sanity check ($XPE > 1$ indicates a measurable gain over the baseline). The narrow spread in C_λ across candidates reflects selection stability: small modeling changes do not flip the ranking.

Case II (RTD): worldview comparison with matched complexity. For RTD, candidates have matched parameter counts, so ranking mainly reflects which *distribution family* best generalizes across runs. Figures 4–5 show that `gamma_2p` provides the best cross-run extrapolation and the lowest C_λ . In this setting, RIEC-L1 preserves the standard engineering conclusion (a tanks-in-series/gamma RTD) while still producing a reusable evidence record. The fitted gamma parameters map directly to mean residence time and dispersion (or an equivalent number of tanks), supporting interpretability for process engineers.

Case III (Drying, LOCO): conservative choice under hard extrapolation. Drying under LOCO split is intentionally challenging: each held-out fold removes an entire (material, temperature) condition, so flexible models can overfit small idiosyncrasies. Figures 6–7 highlight the key disagreement: CV prefers `midilli_4p`, while BIC prefers a simpler 2-parameter family. RIEC-L1 selects `page_2p` because its out-of-fold gain is substantial (high XPE) yet does not require the full flexibility of a 4-parameter form. In this case, `page_2p` and `weibull_2p` are essentially tied in evidence (Supplementary S1 ledger), so the specific 2-parameter representative is chosen by a deterministic tie-break; the substantive distinction is the conservative preference for 2 parameters over 4 under hard extrapolation. Quantitatively, `midilli_4p` lowers the CV risk from 4.15×10^{-3} (`page_2p`) to 4.00×10^{-3} (about a 3.5% improvement), but it doubles the parameter count (4 vs 2) and is penalized by the structural term (BIC increases by $\Delta BIC \approx 4.6$ relative to `page_2p`). RIEC-L1 therefore accepts a modest loss in CV fit to halve model complexity and improve robustness and interpretability under hard out-of-condition deployment. This illustrates the intended policy of RIEC: *do not add complexity unless the out-of-fold evidence is clearly worth it.*

Table 1

Three-case summary. Lower is better for BIC and C_λ . XPE > 1 indicates out-of-fold improvement over baseline. In the drying case, page_2p and weibull_2p are nearly tied at the reported precision; the reported winners follow a deterministic tie-break (see the full ledger in Supplementary S1).

Case	RIEC best	BIC best	CV best	n_{eff}	λ	XPE (RIEC)
Optics	cos_poly_deg2	cos_poly_deg2	cos_poly_deg2	29872	0.097	1.035
RTD	gamma_2p	gamma_2p	gamma_2p	10401	0.108	1.000
Drying	page_2p	weibull_2p	midilli_4p	72	0.234	1.569

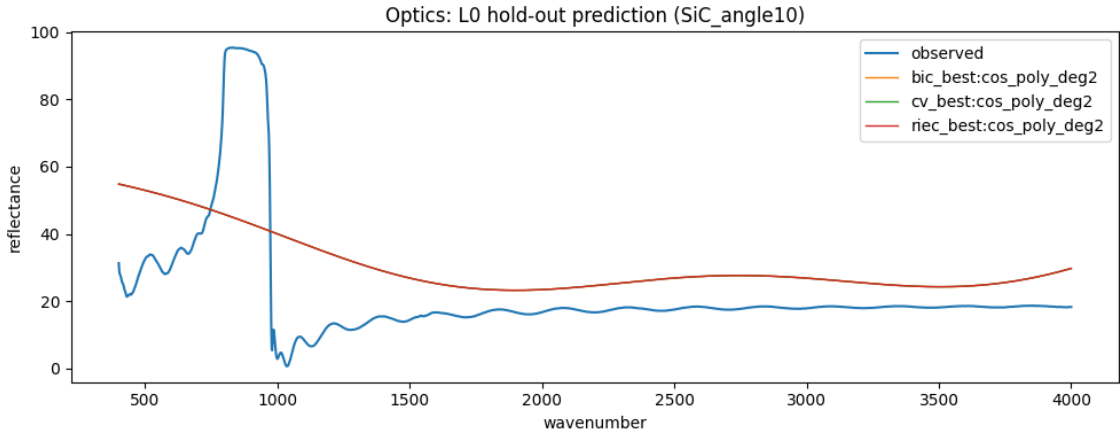


Figure 2: Case I (Optics): holdout overlay. Leave-one-spectrum-out holdout overlay for an unseen spectrum (SiC at 10° incidence; representative condition). The blue curve is the observed reflectance $R(w)$ on the held-out spectrum; the colored curves are predictions from the models chosen by BIC, task-aligned CV, and RIEC-L1. In this large- n_{eff} case the three selectors coincide on cos_poly_deg2, hence the predicted curves overlap. The overlay is an auditable check that the selected complexity captures the main spectral trend without introducing visually obvious overfitting artifacts.

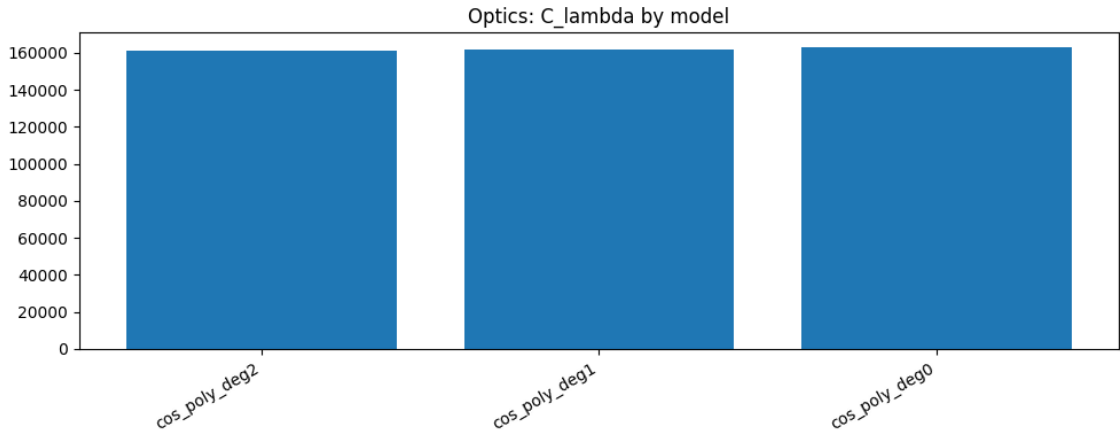


Figure 3: Case I (Optics): C_λ ranking evidence. Candidate ranking by the hybrid score C_λ across the proxy library $\{\text{cos_poly_deg0/1/2}\}$ (lower is better). The minimum at cos_poly_deg2 matches the selection in Table 1 and is supported by a small but consistent out-of-fold improvement (XPE > 1) while the BIC penalty remains modest. The tight spread of scores indicates selection stability in the large- n_{eff} regime.

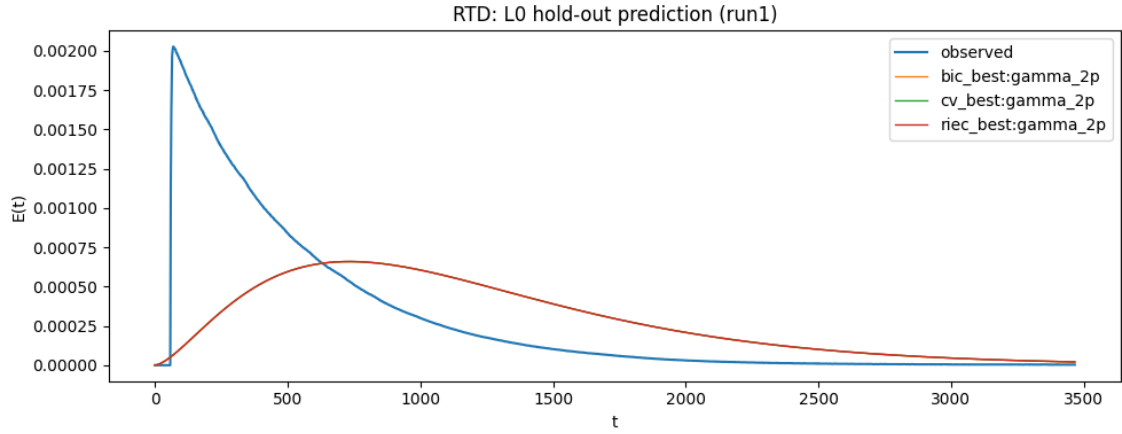


Figure 4: Case II (RTD): holdout overlay. Leave-one-run-out holdout overlay for an unseen tracer-response run (representative example). We plot the empirical RTD kernel $E(t)$ (blue) together with predictions from the models selected by BIC, CV, and RIEC-L1. In this moderate- n_{eff} case all three selectors choose gamma_2p , hence the predicted curves coincide. This demonstrates that the hybrid criterion preserves the engineering conclusion when evidence is strong and the candidate worldviews are well separated.

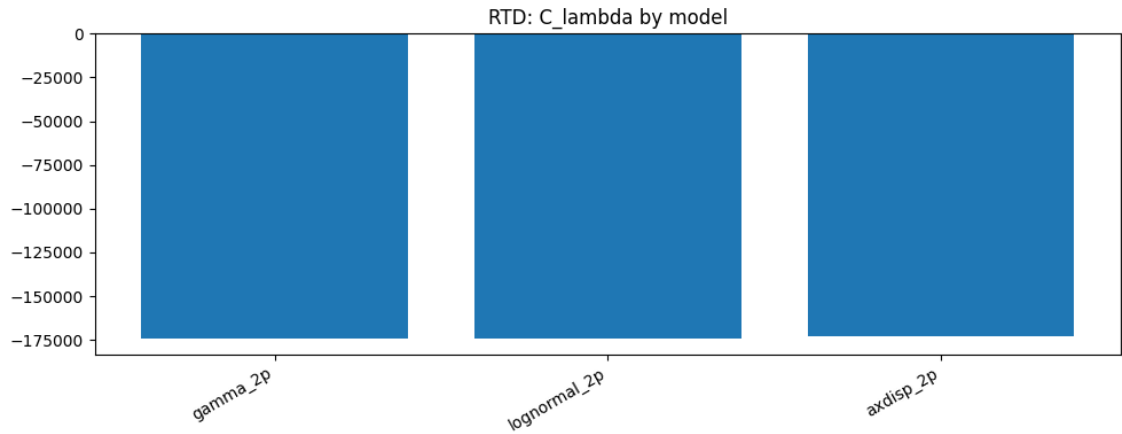


Figure 5: Case II (RTD): C_λ ranking evidence. Candidate ranking by C_λ over three competing RTD families (gamma_2p , lognormal_2p , axdisp_2p). Because parameter counts are matched, this ranking primarily reflects worldview adequacy under out-of-fold evaluation. gamma_2p achieves the lowest score, consistent with its interpretable tanks-in-series form and stable generalization across runs.

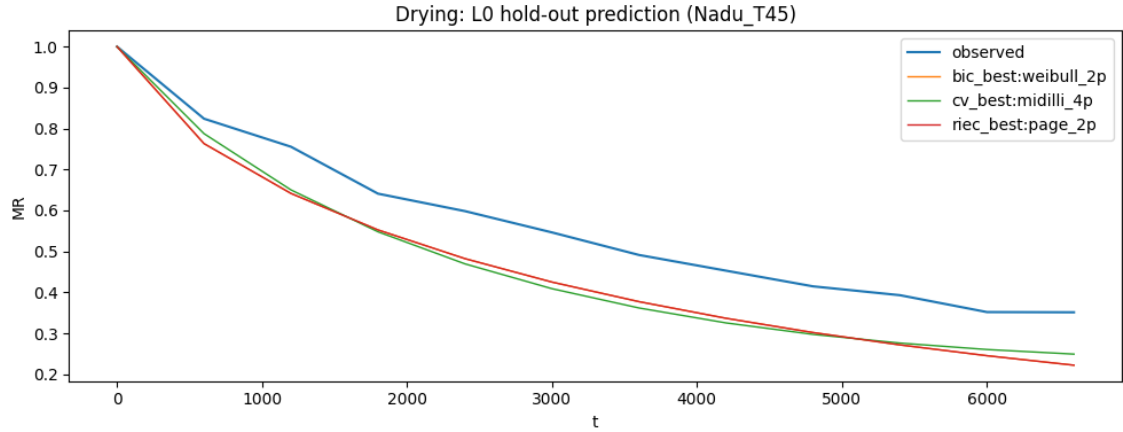


Figure 6: Case III (Drying): holdout overlay under LOCO extrapolation. Leave-one-condition-out (material, temperature) holdout overlay for an unseen condition (Nadu at 45°C; representative). The flexible Midilli model achieves the lowest CV error (midilli_4p, green), whereas BIC favors a different 2-parameter family (weibull_2p, orange); RIEC-L1 selects the 2-parameter Page model (page_2p, red) as a robustness-oriented compromise. Visually, the RIEC choice tracks the holdout decay curve well while avoiding the extra degrees of freedom that may overfit in small-sample extrapolation.

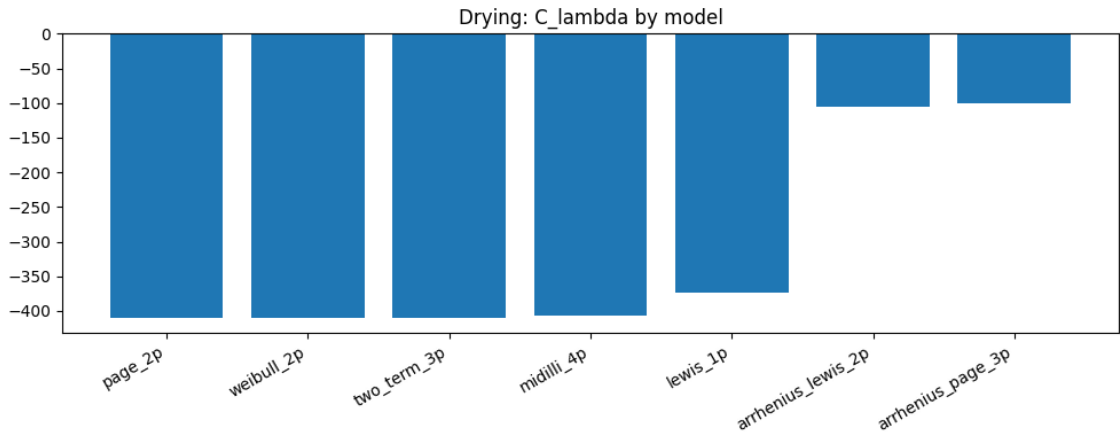


Figure 7: Case III (Drying): C_λ ranking evidence. Candidate ranking by C_λ across drying model families (lower is better). The hybrid score penalizes additional flexibility (e.g., midilli_4p) unless its out-of-fold gains are large enough to outweigh complexity, leading to the selection of page_2p. Together with Fig. 6, this ranking illustrates how RIEC-L1 operationalizes a conservative “do not add complexity unless necessary” policy under LOCO extrapolation.

6. Discussion

Across three cases, we address a practical question: in industrial curve tasks, is “more complex” truly necessary, and how can we decide using auditable evidence? When sample sizes are large or structural worldviews are clearly separated (Optics, RTD), RIEC agrees with BIC/CV and does not change the engineering conclusion. When sample sizes are small and extrapolation is harder (Drying, LOCO), CV can reward flexible models for marginal holdout gains; RIEC provides a conservative compromise that prioritizes parsimony when improvements are modest. In deployment, this reduces the risk of parameter inflation that complicates interpretation, tuning, and transfer to new conditions.

Practical usage in engineering workflows. RIEC is designed to be used as a lightweight “evidence gate” before deployment. In practice, we recommend (i) choosing a split that mirrors deployment (by run, by condition, or by batch), (ii) reporting the full ranking ledger (not only the winner), and (iii) inspecting at least one holdout overlay per group as a human-auditable check. When BIC and CV agree, the decision is typically stable; when they disagree, the ranking table provides a concrete starting point for engineering discussion (e.g., whether interpretability, stability, or accuracy is the dominant requirement). Beyond the three case studies here, the same protocol applies to many curve-like industrial tasks, including chemical process temperature–pressure trajectories, battery charge/discharge curves, motor vibration envelopes, and pharmaceutical dissolution profiles, where deployment often involves extrapolation across batches or operating conditions.

Limitations and future work. This paper intentionally uses a minimal Layer-1 instantiation. First, n_{eff} is treated as a transparent scalar-sample count to balance engineering usability and theoretical rigor; in curve-fitting the residuals can be serially correlated, so more refined effective-sample corrections (e.g., block-based or correlation-adjusted n_{eff}) may be beneficial in strongly dependent settings. Importantly, RIEC-L1 is compatible with any alternative n_{eff} definition because it only enters through $\log(n_{\text{eff}})$. Second, the current $\lambda(n_{\text{eff}}) = 1/\log n_{\text{eff}}$ is a simple monotone schedule; future layers could learn or calibrate λ for specific application families, or incorporate domain priors (e.g., physical constraints) as additional evidence channels. Finally, we focus on small model libraries; extending the approach to large hypothesis spaces (e.g., automated basis construction) is a natural direction.

7. Conclusion

We implement RIEC-L1 for industrial curve-fitting tasks: over a finite model library $M = (C, f)$, the hybrid score C_λ unifies BIC-style penalties with out-of-fold evidence (CV risk and XPE) into one auditable ranking table. For optical spectra and RTD, RIEC agrees with BIC/CV. For thin-layer drying, RIEC moderates the tendency of CV to favor the most flexible model in small-sample extrapolation. In industrial deployment, this can reduce overfitting-driven extrapolation failures, lower model maintenance cost, and provide a traceable model-selection record that supports engineering review and quality management. Supplementary materials provide a reader-oriented guide (S0), a one-click reproducibility protocol plus materials and an output snapshot for quick inspection (S1), formal definitions and derivations (S2), and domain background with engineering linkage notes (S3).

Declarations

CRedit authorship contribution statement

Jincheng Li: Conceptualization; Methodology; Software; Formal analysis; Visualization; Writing – original draft. Yuchen Zhao: Writing – review & editing (domain-specific mathematical modeling checks). Xifeng Li: Resources; Writing – review & editing (domain-specific mathematical modeling checks).

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare no competing interests.

Data and code availability

For peer review, all materials needed to reproduce the experiments, figures, and tables are provided in Supplementary S1 (data, scripts, and an output snapshot for quick inspection without running code). Upon acceptance, a public, versioned repository release will be made available under the corresponding author's GitHub account <https://github.com/lijincheng494gmail>. Supplementary S2–S3 provide formal definitions/derivations and domain background notes, respectively.

Declaration of generative AI and AI-assisted technologies

Generative AI tools were used for English-language proofreading and to assist in preparing the final submission files (e.g., formatting and consistency checks). All scientific content, analyses, and conclusions were produced and verified by the authors.

References

- [1] G. E. Schwarz, Estimating the dimension of a model, *Annals of Statistics* 6 (2) (1978) 461–464. doi:10.1214/aos/1176344136.
- [2] M. Stone, Cross-validatory choice and assessment of statistical predictions, *Journal of the Royal Statistical Society: Series B (Methodological)* 36 (2) (1974) 111–147.
- [3] S. Arlot, A. Celisse, A survey of cross-validation procedures for model selection, *Statistics Surveys* 4 (2010) 40–79. doi:10.1214/09-SS054.
- [4] C. Maduwanthi, Chathurika maduwanthi (dataset), Mendeley Data, V1 (2021). doi:10.17632/jdbkxgn2v5.1.
- [5] A. Midilli, H. Kucuk, Z. Yapar, A new model for single-layer drying, *Drying Technology* 20 (7) (2002) 1503–1513. doi:10.1081/DRT-120005864.
- [6] G. E. Page, Factors influencing the maximum rates of air drying shelled corn in thin layers, M.s. thesis, Purdue University (1949).
- [7] M. Stone, An asymptotic equivalence of choice of model by cross-validation and Akaike's criterion, *Journal of the Royal Statistical Society: Series B (Methodological)* 39 (1) (1977) 44–47.
- [8] S. Watanabe, Asymptotic equivalence of Bayes cross validation and widely applicable information criterion in singular learning theory, *Journal of Machine Learning Research* 11 (2010) 3571–3594. URL <https://www.jmlr.org/papers/v11/watanabe10a.html>.
- [9] A. Vehtari, A. Gelman, J. Gabry, Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC, *Statistics and Computing* 27 (5) (2017) 1413–1432. doi:10.1007/s11222-016-9696-4.
- [10] J. Shao, Linear model selection by cross-validation, *Journal of the American Statistical Association* 88 (422) (1993) 486–494. doi:10.1080/01621459.1993.10476299.
- [11] P. V. Danckwerts, Continuous flow systems: Distribution of residence times, *Chemical Engineering Science* 2 (1) (1953) 1–13. doi:10.1016/0009-2509(53)80001-1.
- [12] O. Levenspiel, *Chemical Reaction Engineering*, 3rd Edition, Wiley, New York, 1999.
- [13] M. C. Tropicovsky, A. S. Sabau, A. R. Lupini, Z. Zhang, Transfer-matrix formalism for the calculation of optical response in multilayer systems: from coherent to incoherent interference, *Optics Express* 18 (24) (2010) 24715–24721. doi:10.1364/OE.18.024715.
- [14] O. S. Heavens, *Optical Properties of Thin Solid Films*, Dover Publications, New York, 1991.
- [15] O. Levenspiel, W. K. Smith, Notes on the diffusion-type model for the longitudinal mixing of fluids in flow, *Chemical Engineering Science* 6 (4–5) (1957) 227–233. doi:10.1016/0009-2509(57)85021-0.
- [16] S. M. Henderson, S. Pabis, Grain drying theory i: Temperature effect on drying coefficient, *Journal of Agricultural Engineering Research* 6 (1961) 169–174.

[Click here to view linked References](#)

RIEC-L1 (Results in Engineering) – LaTeX editable source package

Contents

- main.tex : manuscript source (CAS single-column style: cas-sc)
- main.bbl : reference list (already compiled; main.tex uses `\input{main.bbl}`)
- references.bib : BibTeX database (provided for editability)
- cas-sc.cls, cas-common.sty : Elsevier CAS class/style (included for safety)
- figures/ : external figure files referenced by main.tex (Figures 2–7)

How to compile locally (example)

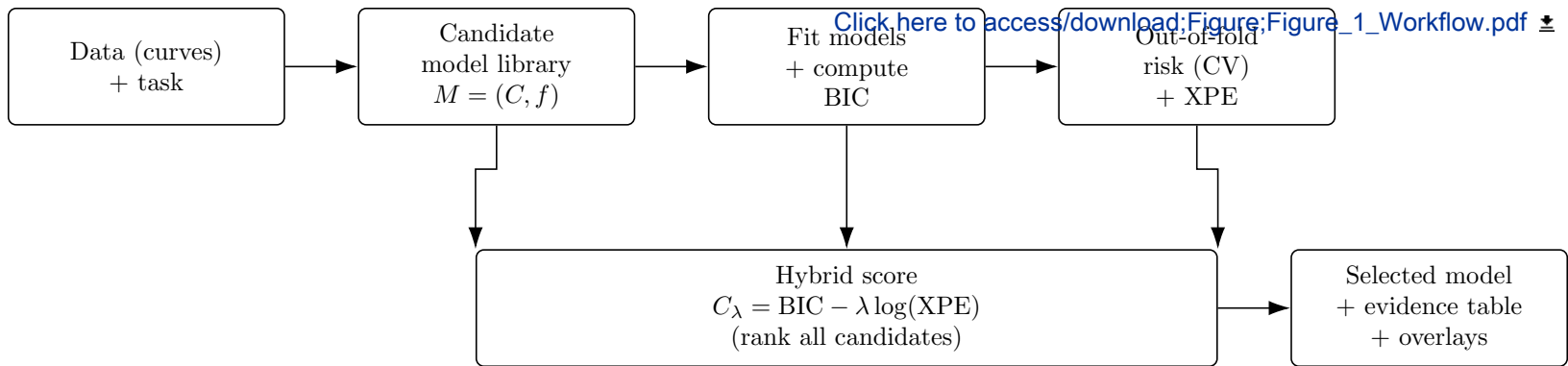
- 1) Ensure a TeX distribution is installed (TeX Live / MikTeX).
- 2) From this folder, run:

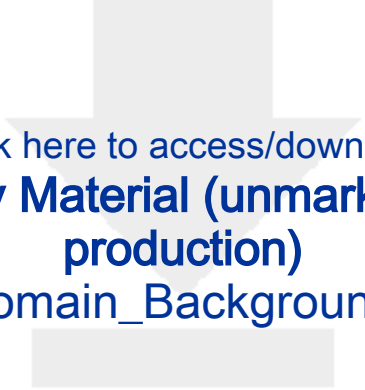
```
pdflatex main.tex
pdflatex main.tex
```

Because main.tex inputs main.bbl directly, BibTeX is not required for a clean build.

Notes

- Figure 1 (workflow) is drawn directly in main.tex using TikZ.
- If you prefer BibTeX, replace "`\input{main.bbl}`" with a standard bibliography call (e.g., `\bibliographystyle{elsarticle-num}` and `\bibliography{references}`).

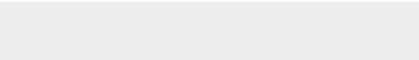
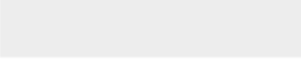


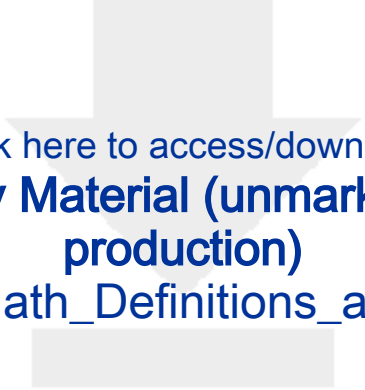


Click here to access/download

**Supplementary Material (unmarked version, for
production)**

Supplement_S3_Domain_Background_and_Linkage.pdf

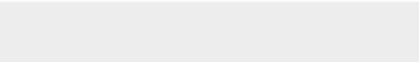
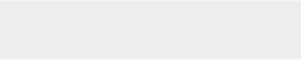


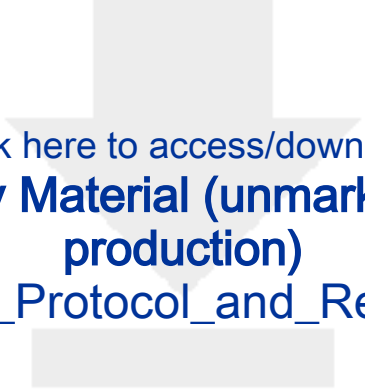


Click here to access/download

**Supplementary Material (unmarked version, for
production)**

Supplement_S2_Math_Definitions_and_Derivations.pdf

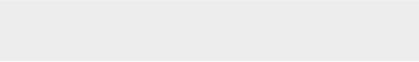
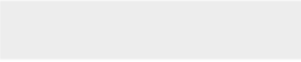


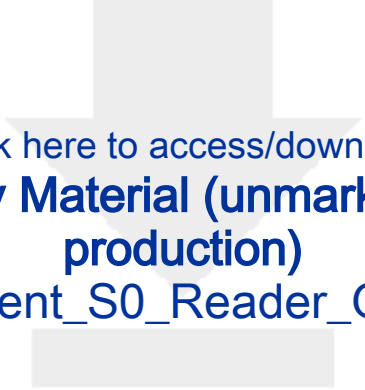


[Click here to access/download](#)

**Supplementary Material (unmarked version, for
production)**

[Supplement_S1_Protocol_and_Reproducibility.pdf](#)

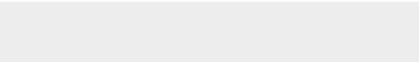
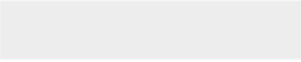


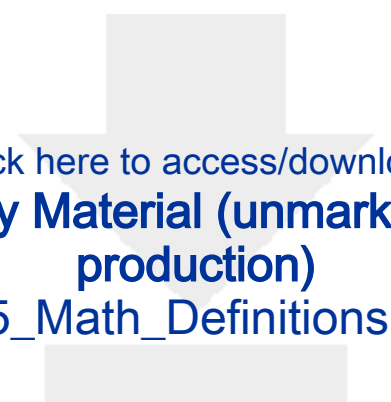


Click here to access/download

**Supplementary Material (unmarked version, for
production)**

Supplement_S0_Reader_Guide.pdf

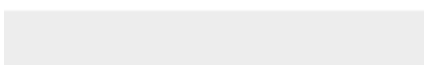


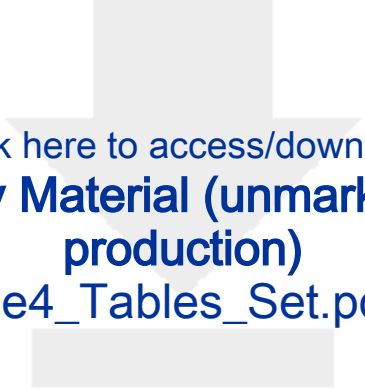


Click here to access/download

**Supplementary Material (unmarked version, for
production)**

file5_Math_Definitions.pdf

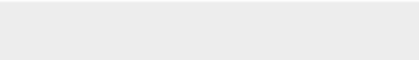
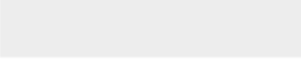


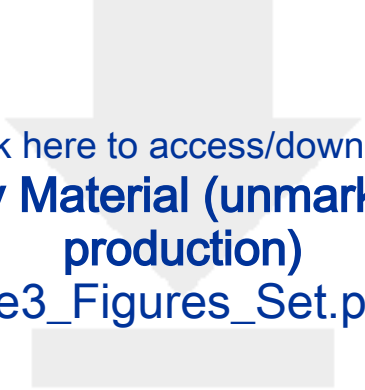


Click here to access/download

**Supplementary Material (unmarked version, for
production)**

file4_Tables_Set.pdf

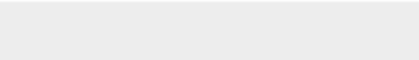
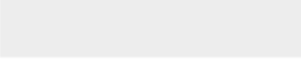


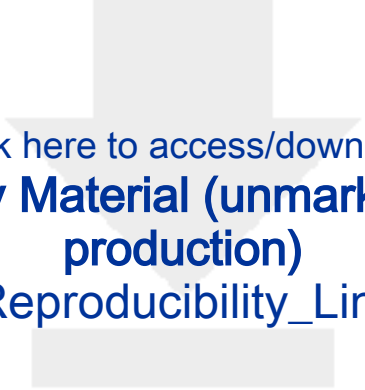


Click here to access/download

**Supplementary Material (unmarked version, for
production)**

file3_Figures_Set.pdf





Click here to access/download

**Supplementary Material (unmarked version, for
production)**

file7_Reproducibility_Links.pdf

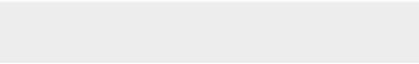


Table 1. Three-case summary.

Lower is better for BIC and $C\lambda$. $XPE > 1$ indicates out-of-fold improvement over baseline. In the drying case, `page_2p` and `weibull_2p` are nearly tied at the reported precision; the reported winners follow a deterministic tie-break (see the full ledger in Supplementary S1).

Case	RIEC best	BIC best	CV best	n_eff	λ	XPE (RIEC)
Optics	<code>cos_poly_deg2</code>	<code>cos_poly_deg2</code>	<code>cos_poly_deg2</code>	29872	0.097	1.035
RTD	<code>gamma_2p</code>	<code>gamma_2p</code>	<code>gamma_2p</code>	10401	0.108	1.000
Drying	<code>page_2p</code>	<code>weibull_2p</code>	<code>midilli_4p</code>	72	0.234	1.569



Click here to download Research Data
<https://github.com/lijincheng494gmail/result-in-eg-reproduce>



Robustness-Integrated Evidence and Choice (RIEC)-L1: an auditable hybrid criterion for model selection in industrial curve-fitting tasks (optical spectra, RTD, and thin-layer drying)

Jincheng Li^{a,*}; Yuchen Zhao^b; Xifeng Li^c

^a Faculty of Science and Technology, University of Nottingham Ningbo China (UNNC), 199 Taikang East Road, Yinzhou District, Ningbo, Zhejiang 315100, China

^b Department of Chemistry, College of Chemistry and Chemical Engineering, Xiamen University, 422 Siming Road, Siming District, Xiamen, Fujian 361005, China

^c IT Department, China National Cereals, Oils and Foodstuffs Corporation (COFCO), COFCO Fortune Plaza, No. 8 Chaoyangmen South Street, Chaoyang District, Beijing 100020, China

Corresponding author: Jincheng Li (email: smyjl12@nottingham.edu.cn).